One-Page Cheatsheet: When the Quadratic Approximation Breaks Down

Core Idea:

The update rule delta = -P_t * g_t assumes the loss is well-approximated locally by a quadratic:
f(theta + delta) ≈ f(theta) + g■ delta + 1/2 delta■ H delta
This only works if curvature (the Hessian) changes slowly in a small neighborhood.

Failure Condition:

The quadratic model fails when the Hessian is not stable:
When H(theta + δ) differs sharply from H(theta) even for tiny δ.
This means third derivatives are large and the bowl shape bends unpredictably.

Why This Breaks Optimizers:

- H■^{1} g becomes unreliable.
- All P_t approximations (SGD, Adam, K-FAC, Shampoo) become misleading.
- Even simple SGD can overshoot because curvature varies too quickly.

Mnemonic:

“Quadratic ≈ Bowl. If the bowl twists fast, curvature math collapses.”

Practical Red Flags:

- Sharp jagged loss landscape.
- Very large learning rates.
- Adam/RMSprop variance estimates jump around.
- Second-order methods diverge immediately.
- Training unstable even with small steps.

Key Condition to Remember:

Quadratic approximation ≈ good only when the Hessian is Lipschitz (locally smooth).
Fast-changing curvature kills quadratic reasoning.

20■Second Summary:

For quadratic modeling and linear-transform optimizers to work, curvature must be stable.
If H changes too fast → higher-order terms dominate → the quadratic view breaks.